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DAY 5 · PROJECT 1

AI Pipelines for Document Analysis

From Text to Structured Insight

Building the Resume × Job Description Analyzer

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By the End of This Morning

1. Explain why one prompt can't do a recruiter's job — and design a pipeline that can.
2. Turn a résumé and a job post into structured JSON your code can trust.
3. Score a match the way an ATS does — and explain exactly why.

This is not prompt engineering, round two. *It's software engineering — applied to prompts.*

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This Morning

1. **One Prompt Isn't Enough**
   Why chatbots fail at analysis
2. **The Assembly Line**
   Parse → Extract → Analyse → Score
3. **Unstructured → Structured**
   JSON as a contract between stages
4. **Prompts That Judge, Not Just Chat**
   Extraction vs. evaluation vs. feedback
5. **Trust the Number**
   Scoring, thresholds, and when the model misbehaves

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SECTION 01 OF 05

One Prompt Isn't Enough

01

Why asking a chatbot to extract, evaluate, score, and advise all at once backfires.

The Recruiter's Afternoon

200

résumés, one afternoon

"Read this résumé, decide if the skills match, give me a percentage, and explain what's missing — all in one response."

Inconsistent format

A different shape every time — nothing to compare across candidates.

Possible hallucination

The model may invent a skill the applicant never listed.

Not comparable

Free-form prose can't be ranked, filtered, or stored.

Three Reasons One Call Fails

1

Attention dilution

Multiple subtasks at once means less quality on each one.

2

Format conflict

Tight JSON and flowing prose fight inside one response.

3

No intermediate check

A bad extraction poisons everything built on top of it.

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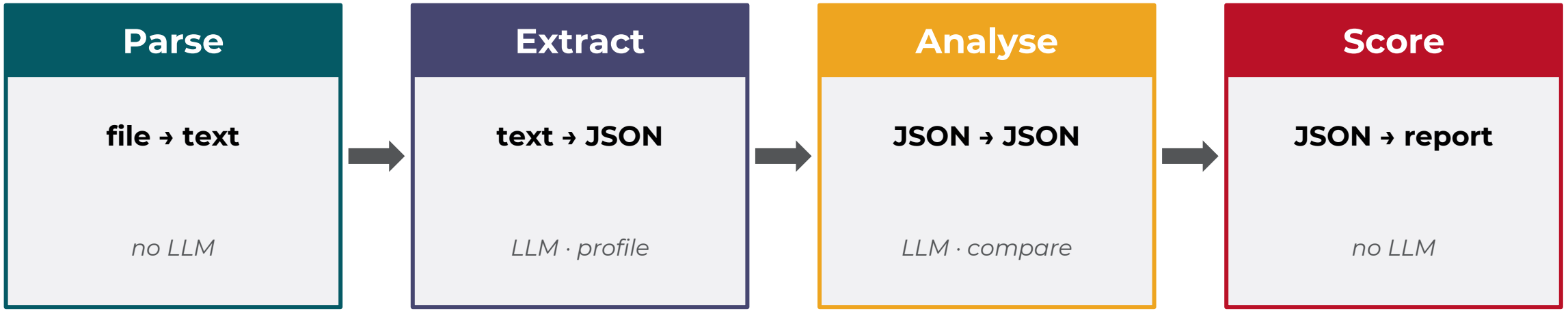
SECTION 02 OF 05

The Assembly Line

02

Each station does exactly one job — and you can test it without the others existing.

One Job Per Station



Data flows left to right. No stage looks back — and none knows what any other stage does internally.

Every Step Reduces Ambiguity

Stage Boundary	Before	After
File → Text	`resume.pdf (binary bytes)`	`"Aiko Tanaka … Skills: Python, AWS"`
Text → JSON	`"…proficient in Python, cloud projects…"`	`{ skills: { languages: [Python] } }`
JSON → Analysis	`profile JSON + requirements JSON`	`{ present, missing, score: 72 }`
Analysis → Report	`{ score: 72, missing: [Docker] }`	`"ATS score 72/100. Bonus: Docker."`

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SECTION 03 OF 05

Unstructured → Structured

03

Prose says a lot; a computer can't act on it. JSON is the bridge — and the contract.

The LLM Is a Translator

UNSTRUCTURED TEXT

```
Experienced in Python and cloud
infrastructure. Led a team of 3.
Familiar with AWS, Docker, and
some Kubernetes.
```

LLM

STRUCTURED JSON

```
{
  "languages": ["Python"],
  "tools": ["AWS", "Docker"],
  "leadership": true
}
```

A dict lookup can't read prose. An LLM reads prose and writes dictionaries — that's the bridge.

The Schema Is a Contract

Rename one field `"missing" → "gaps"` in Stage 3.

Stage 4 reads it with `result.get("missing", [])`.

> ***No KeyError. No crash. Just a report that silently claims nothing is missing — when something is.***

Schema-first design: decide the exact JSON shape before writing the prompt. In ICCO terms, the schema *is* the Output component.

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SECTION 04 OF 05

Prompts That Judge

Not just chat: extraction copies facts, evaluation judges them, feedback explains them. Never confuse the three.

Three Prompts, Three Jobs

Extraction

Copy facts from text into a schema.

temp 0.0 – 0.1

Only extract. Never invent.

Evaluation

Compare two structured objects against a rubric.

temp 0.1 – 0.3

Cite the rubric. Score strictly.

Feedback

Translate results into actionable advice.

temp 0.3 – 0.6

Feedback only. Never rewrite.

Temperature: A Precision Dial

Fully deterministic → more nuance, still parseable → unreliable for JSON

0.0 – 0.1	0.1 – 0.3	0.3 – 0.6	0.7+
Extraction	Evaluation	Feedback	Avoid for JSON

Above 0.7, JSON-returning prompts risk malformed structure. Extraction stays near zero — it's copying, not creating.

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SECTION 05 OF 05

Trust the Number

A composite score only means something next to a threshold — and a way to explain it.

Weighted Aggregation

Dimension	Weight	Score
Keyword match	0.5	80
Bullet quality	0.3	65
Structure audit	0.2	70

$$S = \Sigma\, w_i \cdot s_i$$ **(weights sum to 1.0)**

73.5

out of 100 — composite score

Weights encode the evaluator's priorities. Choose them deliberately — and document them.

The Threshold Makes the Score Mean Something

```
{
  "composite_score": 64,
  "pass_threshold": 60,
  "result": "PASS",
  "margin": 4
}
```

PASS

64 clears a **60** threshold by a margin of **+4**. Against a 70% bar, the same résumé fails.

Singapore ATS thresholds typically sit at 60–70% for entry-level tech roles — a heuristic, not a law. Report the score, the threshold, and the margin together.

The LLM Is Not a Database

Leading prose

"Here is the JSON you requested:"

Markdown fences

```json ... ``` wrapped around the object

Wrong types

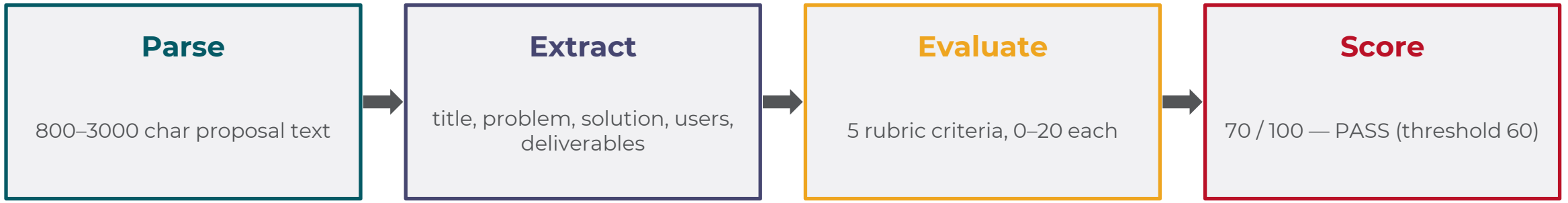
An array field returns as a bare "string"

Truncation

Response hits max_tokens mid-object

Never return None or {} on a parse failure. Raise — and let the caller retry.
```

Worked Example: The Proposal Analyser




```
"user_specificity": { "score": 10, "note": "Target users described only as ‘students’ — too broad." }
```

An observation the student can act on — not a rewritten user-persona paragraph handed to them.

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CONTRIBUTIONS

What You Can Now Do

- ✓ Explain why pipelines beat a single prompt — and design one that separates parse, extract, analyse, and score.
- ✓ Write a schema-first extraction prompt and a rubric-grounded evaluation prompt, at the right temperature.
- ✓ Compute a weighted, threshold-checked composite score — and defend every number in it.

This afternoon: build the Resume × Job Description Analyzer.

Questions?